Reed Maxwell, Professor

Wendy Zhou, Professor

Department of Geophysics

John Bradford, Vice President for Global Initiatives

Brandon Dugan, Professor and Baker Hughes Chair in Petrophysics & Borehole Geophysics and Associate Department Head GP

Matthew Siegfried, Assistant Professor

Department of Humanities, Arts and Social Sciences

Hussein Amery, Professor

Adrianne Kroepsch, Assistant Professor

Department of Mechanical Engineering

Nils Tilton, Associate Professor

Department of Mining Engineering

Rennie Kaunda, Assistant Professor

Department of Petroleum Engineering

Yu-Shu Wu, Professor

Materials Science

Degrees Offered

- Master of Science (Materials Science, thesis option or non-thesis option)
- Doctor of Philosophy (Materials Science)

Program Description

The interdisciplinary graduate program in Materials Science exists to educate students, with at least a Bachelor of Science degree in engineering or science in the diverse field of Materials Science. This diversity includes the four key foundational aspects of Materials Science – materials properties including characterization and modeling, materials structures, materials synthesis and processing, and materials performance – as applied to materials of a variety of types (i.e., metals, ceramics, polymers, electronic materials and biomaterials). The Materials Science graduate program is responsible for administering MS (thesis and non-thesis) and PhD degrees in Materials Science.

This interdisciplinary degree program coexists alongside strong disciplinary programs in Chemistry, Chemical and Biochemical Engineering, Mechanical Engineering, Metallurgical and Materials Engineering, Mining, and Physics. The student's graduate committee will have final approval of the course of study.

Fields of Research

- Advanced polymeric materials
- Alloy theory, concurrent design, theory-assisted materials engineering, and electronic structure theory
- Applications of artificial intelligence techniques to materials processing and manufacturing, neural networks for process modeling and sensor data processing, manufacturing process control
- Atomic scale characterization
- Atom Probe Tomography
- Biomaterials
- Ceramic processing, modeling of ceramic processing
- Characterization, thermal stability, and thermal degradation mechanisms of polymers
- Chemical and physical processing of materials, engineered materials, materials synthesis
- Chemical vapor deposition
- Coating materials and applications
- Computational condensed matter physics, semiconductor alloys, first-principles phonon calculations
- Computer modeling and simulation
- Control systems engineering, artificial neural systems for senior data processing, polymer cure monitoring sensors, process monitoring and control for composites manufacturing
- Crystal and molecular structure determination by X-ray crystallography
- Electrodeposition
- Electron and ion microscopy
- Energetic materials (explosives) and processing
- Energy storage
- Experimental condensed-matter physics, thermal and electrical properties of materials, superconductivity, photovoltaics
- Fuel cell materials
- Fullerene synthesis, combustion chemistry
- Heterogeneous catalysis, reformulated and alcohol fuels, surface analysis, electrophotography
- High-temperature ceramics
- Intelligent automated systems, intelligent process control, robotics, artificial neural systems
- Materials synthesis, interfaces, flocculation, fine particles
- Mathematical modeling of material processes
- Mechanical metallurgy, failure analysis, deformation of materials, advanced steel coatings
- Mechanical properties of ceramics and ceramic composites
- High entropy alloys
- Mössbauer spectroscopy, ion implantation, small-angle X-ray scattering, semiconductor defects
- Nano materials
- Nondestructive evaluation
- Nonferrous structural alloys
- Novel separation processes: membranes, catalytic membrane reactors, biopolymer adsorbents for heavy metal remediation of ground surface water
- Numerical modeling of particulate media, thermomechanical analysis
- Optical properties of materials and interfaces
- Phase transformations and mechanisms of microstructural change
- Photovoltaic materials and device processing
- Physical metallurgy, ferrous and nonferrous alloy systems
- Physical vapor deposition, thin films, coatings
- Power electronics, plasma physics, pulsed power, plasma material processing

- Processing and characterization of electroceramics (ferro-electrics, piezoelectrics, pyroelectrics, and dielectrics)
- Semiconductor materials and device processing
- Soft materials
- Solidification and near net shape processing
- Surface physics, epitaxial growth, interfacial science, adsorption
- Thermoelectric materials
- Transport phenomena and mathematical modeling
- Weld metallurgy, materials joining processes
- Welding and joining science

Combined Degree Option

Mines undergraduate students have the opportunity to begin work on an MS non-thesis degree while concurrently completing their Bachelor's degree at Mines.

Dual-Degree Program Option

Students have the opportunity to earn two degrees with the dual degree option. Students complete coursework to satisfy requirements for both a non-thesis MS in Materials Science from Colorado School of Mines and an MS of Physical Chemistry and Chemical Physics from the University of Bordeaux.

Primary Contact

Materials Science Program
https://materials.mines.edu/

Program Requirements

Each of the three degree programs (non-thesis MS, thesis-based MS, and PhD) require the successful completion of three core courses for a total of 9 credits that will be applied to the degree program course requirements. Depending upon the individual student's background, waivers for these courses may be approved by the program director. In order to gain a truly interdisciplinary understanding of Materials Science, students in the program are encouraged to select elective courses from several different departments outside of their home department regardless of the existence of a cross-listed MLGN course number. Course selection should be completed in consultation with the student's advisor and/or program director as appropriate.

Mines' Combined Undergraduate / Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to six credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with "B-" or better, not be substitutes for required coursework, and meet all other University, Department, and Program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

Listed below are the three required Materials Science core courses:

MLGN591	MATERIALS THERMODYNAMICS	3.0
MLGN592	ADVANCED MATERIALS KINETICS AND TRANSPORT	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
Total Semester Hrs		**9.0**

Master of Science (Thesis Option)

The Master of Science degree requires a minimum of 30.0 credits of acceptable coursework and thesis research credits (see table below). The student must also submit a thesis and pass the Defense of Thesis examination before the Thesis Committee.

MLGN591	MATERIALS THERMODYNAMICS	3.0
MLGN592	ADVANCED MATERIALS KINETICS AND TRANSPORT	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
ELECTIVES	Materials Science or Related Courses in any participating Materials Science department (AMFG, CBEN, CEEN, CHGN, DSCI, MEGN, MTGN)	9.0
MLGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT	12.0
Total Semester Hrs		**30.0**

Master of Science (Non-Thesis Option with a case study)

The Master of Science degree requires a minimum of 24.0 credits of acceptable course work and 6.0 additional credits of either case study or designated design course including:

MLGN591	MATERIALS THERMODYNAMICS	3.0
MLGN592	ADVANCED MATERIALS KINETICS AND TRANSPORT	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
MLGN	Case Study OR Designated Design Courses *See table below	6.0
ELECTIVES	Materials Science or Related Courses in any participating Materials Science department (AMFG, CBEN, CEEN, CHGN, DSCI, MEGN, MTGN)	15.0
Total Semester Hrs		**30.0**

*MLGN Design Courses

MLGN500	PROCESSING, MICROSTRUCTURE, AND PROPERTIES OF MATERIALS	3.0
MLGN505	MECHANICAL PROPERTIES OF MATERIALS	3.0
MLGN510	SURFACE CHEMISTRY	3.0
MLGN516	PROPERTIES OF CERAMICS	3.0
MLGN535	INTERDISCIPLINARY MICROELECTRONICS PROCESSING LABORATORY	3.0
MLGN544	ADVANCED PROCESSING OF CERAMICS	3.0
MLGN561	TRANSPORT PHENOMENA IN MATERIALS PROCESSING	3.0
MLGN569	FUEL CELL SCIENCE AND TECHNOLOGY	3.0
MLGN583	PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES	3.0

MLGN	Additional 500-level courses can be approved by MLGN Program Director	3.0

Doctor of Philosophy

The Doctor of Philosophy degree requires a minimum of 72.0 credits of course and research credit including:

MLGN591	MATERIALS THERMODYNAMICS	3.0
MLGN592	ADVANCED MATERIALS KINETICS AND TRANSPORT	3.0
MLGN593	BONDING, STRUCTURE, AND CRYSTALLOGRAPHY	3.0
MLGN707	GRADUATE THESIS / DISSERTATION RESEARCH CREDIT Minimum	24.0
ELECTIVES	Materials Science or Related Courses in any participating Materials Science department (AMFG, CBEN, CEEN, CHGN, DSCI, MEGN, MTGN) Minimum	15.0
MLGN	Thesis Research or Material Science Coursework	24.0
Total Semester Hrs		**72.0**

A minimum of 15 course credits earned at Mines is required for the PhD degree. In exceptional cases, this 15 Mines course credit requirement can be reduced in part or in full through the written consent of the student's advisor, program director, and thesis committee.

PhD Qualifying Process

The following constitutes the qualifying processes by which doctoral students are admitted to candidacy in the Materials Science program.

Core Curriculum – The three required core classes must be completed in the first full academic year for all doctoral candidates. Students must obtain a grade of B or better in each class to be eligible to take the qualifying examination at the end of the succeeding summer semester. Students who do not meet the grade requirements may submit a written appeal to the program director by June 1st, outlining their rationale and providing any supporting documentation for the exemption. The graduate affairs committee will review the appeal on a case-by-case basis and will typically require the student to undertake additional assessments, independent study, or interviews to evidence their mastery of the subject matter. These remediations must be completed by the end of the summer semester following completion of the core classes.

PhD Qualifying Examination – A qualifying examination is given annually at the end of the summer semester under the direction of the Materials Science Graduate Affairs Committee. All first-year Materials Science PhD students are expected to successfully complete the qualifying examination to remain in good standing in the program.

The examination consists of a written report and an oral examination defending the written report. The written report should serve as a proposal for a research endeavor worthy of a single publication and discuss both the background that would motivate such an effort and the specific activities proposed. Students submit their research proposal to a panel of reviewers to include a member of the Materials Science Graduate Affairs Committee and one invited subject matter expert approved by the Materials Science Graduate Affairs Committee. The oral examination will investigate the extent to which the student understands the background, motivation, and proposed work of their proposal.

If a student performs below the expectations of the Materials Science faculty on either component of the qualifying exam, they must revise their report and be re-examined by the end of the ensuing fall semester (typically month 15). Satisfactorily completing the oral exam will allow the student to proceed with the PhD program. Not passing the reexamination will lead to the student being removed from the PhD program. Additional details will be provided to the students by the program director including, exact dates, details expectations, and a rubric for grading. Upon completion of these steps and upon completion of all required coursework, candidates are admitted to candidacy.

PhD Thesis Proposal – While the proposal itself should focus on the central topic of a student's research efforts, during the proposal defense, candidates may expect to receive a wide range of questions from the Committee. This would include all manner of questions directly related to the proposal. Candidates, however, should also expect questions related to the major concept areas of Materials Science within the context of a candidate's research focus.

Following successful completion of coursework and the PhD qualifying process, candidates must also submit a thesis and successfully complete the PhD Defense of Thesis examination before the PhD Thesis Committee.

Courses

MLGN500. PROCESSING, MICROSTRUCTURE, AND PROPERTIES OF MATERIALS. 3.0 Semester Hrs.
(II) A summary of the important relationships between the processing, microstructure, and properties of materials. Topics include electronic structure and bonding, crystal structures, lattice defects and mass transport, glasses, phase transformation, important materials processes, and properties including: mechanical and rheological, electrical conductivity, magnetic, dielectric, optical,
thermal, and chemical. In a given year, one of these topics will be given special emphasis. Another area of emphasis is phase equilibria. Prerequisite: none. 3 hours lecture; 3 semester hours.

MLGN502. SOLID STATE PHYSICS. 3.0 Semester Hrs.
An elementary study of the properties of solids including crystalline structure and its determination, lattice vibrations, electrons in metals, and semiconductors. 3 hours lecture; 3 semester hours. Prerequisite: PH320.

MLGN505. MECHANICAL PROPERTIES OF MATERIALS. 3.0 Semester Hrs.
(I) Mechanical properties and relationships. Plastic deformation of crystalline materials. Relationships of microstructures to mechanical strength. Fracture, creep, and fatigue. Prerequisite: MTGN348. 3 hours lecture; 3 hours lab; 3/4 semester hours. *This is a 3 credit-hour graduate course in the Materials Science Program and a 4 credit-hour undergraduate-course in the MTGN program.

MLGN510. SURFACE CHEMISTRY. 3.0 Semester Hrs.
Introduction to colloid systems, capillarity, surface tension and contact angle, adsorption from solution, micelles and microemulsions, the solid/gas interface, surface analytical techniques, Van Der Waal forces, electrical properties and colloid stability, some specific colloid systems (clays, foams and emulsions). Students enrolled for graduate credit in MLGN510 must complete a special project.

MLGN512. CERAMIC ENGINEERING. 3.0 Semester Hrs.
(II) Application of engineering principles to nonmetallic and ceramic materials. Processing of raw materials and production of ceramic bodies, glazes, glasses, enamels, and cements. Firing processes and reactions

in glass bonded as well as mechanically bonded systems. Prerequisite: MTGN348. 3 hours. lecture; 3 semester hours.

MLGN515. ELECTRICAL PROPERTIES AND APPLICATIONS OF MATERIALS. 3.0 Semester Hrs.
(II) Survey of the electrical properties of materials, and the applications of materials as electrical circuit components. The effects of chemistry, processing, and microstructure on the electrical properties will be discussed, along with functions, performance requirements, and testing methods of materials for each type of circuit component. The general topics covered are conductors, resistors, insulators, capacitors, energy convertors, magnetic materials, and integrated circuits. Prerequisites: PHGN200; MTGN311 or MLGN501; MTGN412/MLGN512,. 3 hours lecture; 3 semester hours.

MLGN516. PROPERTIES OF CERAMICS. 3.0 Semester Hrs.
(II) A survey of the properties of ceramic materials and how these properties are determined by the chemical structure (composition), crystal structure, and the microstructure of crystalline ceramics and glasses. Thermal, optical, and mechanical properties of single-phase and multi-phase ceramics, including composites, are covered. Prerequisites: PHGN200, MTGN311 or MLGN501, MTGN412. 3 semester hours: 3 hours lecture.

MLGN517. THEORY OF ELASTICITY. 3.0 Semester Hrs.
This is a graduate course that builds upon the learning outcomes of Continuum Mechanics course to introduce students the fundamentals of Theory of Elasticity. Introduction is realized through theory development, application examples, and numerical solutions. Learning outcomes from this course would be essential to further studies in visco-elasticity and plasticity. Knowledge from this course will enable students to work on variety of engineering applications in Mechanical, Materials, Aerospace, Civil and related engineering fields. This course is cross-listed with MEGN510.
Course Learning Outcomes

- Recall definitions for indicial notation, transformation rules for tensors, and eigenvalue problems. Tensor algebra and tensor calculus.
- Define, and apply, displacement-strain relationships. Strain measurements using strain gauges and rosettes. Calculate principal strains, maximum shear strain in 3D.
- Establish the definitions, and use, stress tensor, traction vector, normal, and shear tractions. Find stresses at a point on a given plane, principal stresses and max shear stress.
- State the general three-dimensional constitutive law for linear elastic materials. Define material symmetry and the engineering notation stiffness matrix for materials with monoclinic, orthotropic, transversely isotropic, cubic symmetry.
- Define, and apply, the generalized form of Hooke's Law for isotropic materials.
- State, and apply, the field equations for linear isotropic elasticity.
- Write clear and complete boundary condition statement.
- Use the semi-inverse method to find solutions for two dimensional elasticity problems.
- Use the Airy stress function to find solutions for two dimensional elasticity problems.
- Define, and apply, yield theories (von Mises and Tresca) for isotropic solids.
- Use the Prandtl stress function to find solutions for torsional elasticity problems.

MLGN519. NON-CRYSTALLINE MATERIALS. 3.0 Semester Hrs.
(I) An introduction to the principles of glass science and engineering and non-crystalline materials in general. Glass formation, structure, crystallization and properties will be covered, along with a survey of commercial glass compositions, manufacturing processes and applications. Prerequisites: MTGN311 or MLGN501; MLGN512/MTGN412. 3 hours lecture; 3 semester hours.

MLGN530. INTRODUCTION TO POLYMER SCIENCE. 3.0 Semester Hrs.
Chemistry and thermodynamics of polymers and polymer solutions. Reaction engineering of polymerization. Characterization techniques based on solution properties. Materials science of polymers in varying physical states. Processing operations for polymeric materials and use in separations. Prerequisite: CHGN221, MATH225, CHEN357. 3 hour lecture, 3 semester hours.

MLGN531. POLYMER ENGINEERING AND TECHNOLOGY. 3.0 Semester Hrs.
(II) This class provides a background in polymer fluid mechanics, polymer rheological response and polymer shape forming. The class begins with a discussion of the definition and measurement of material properties. Interrelationships among the material response functions are elucidated and relevant correlations between experimental data and material response in real flow situations are given. Processing operations for polymeric materials will then be addressed. These include the flow of polymers through circular, slit, and complex dies. Fiber spinning, film blowing, extrusion and co-extrusion will be covered as will injection molding. Graduate students are required to write a term paper and take separate examinations which are at a more advanced level. Prerequisite: CRGN307, EGGN351 or equivalent. 3 hours lecture; 3 semester hours.

MLGN535. INTERDISCIPLINARY MICROELECTRONICS PROCESSING LABORATORY. 0-3 Semester Hr.
Equivalent with
CBEN435,CBEN535,CHEN435,CHEN535,PHGN435,PHGN535,
(II) Application of science and engineering principles to the design, fabrication, and testing of microelectronic devices. Emphasis on specific unit operations and the interrelation among processing steps. Prerequisite: none. 3 hours lecture; 3 semester hours.

MLGN536. ADVANCED POLYMER SYNTHESIS. 3.0 Semester Hrs.
(II) An advanced course in the synthesis of macromolecules. Various methods of polymerization will be discussed with an emphasis on the specifics concerning the syntheses of different classes of organic and inorganic polymers. Prerequisite: CHGN430, ChEN415, MLGN530. 3 hours lecture, 3 semester hours.

MLGN544. ADVANCED PROCESSING OF CERAMICS. 3.0 Semester Hrs.
A description of the principles of ceramic processing and the relationship between processing and microstructure. Raw materials and raw material preparation, forming and fabrication, thermal processing, and finishing of ceramic materials will be covered. Principles will be illustrated by case studies on specific ceramic materials. A project to design a ceramic fabrication process is required.

MLGN561. TRANSPORT PHENOMENA IN MATERIALS PROCESSING. 3.0 Semester Hrs.
(II) Fluid flow, heat and mass transfer applied to processing of materials. Rheology of polymers, liquid metal/particles slurries, and particulate solids. Transient flow behavior of these materials in various geometries, including infiltration of liquids in porous media. Mixing and blending. Flow behavior of jets, drainage of films and particle fluidization. Surface-tension-, electromagnetic-, and bubble-driven flows. Heat -transfer behavior in porous bodies applied to sintering and solidification of

composites. Simultaneous heat-and-mass-transfer applied to spray drying and drying porous bodies. Prerequisites: ChEN307 or ChEN308 or MTGN461. 3 hours lecture; 3 semester hours.

MLGN563. POLYMER ENGINEERING: STRUCTURE, PROPERTIES AND PROCESSING. 3.0 Semester Hrs.
(II) An introduction to the structure and properties of polymeric materials, their deformation and failure mechanisms, and the design and fabrication of polymeric end items. The molecular and crystallographic structures of polymers will be developed and related to the elastic, viscoelastic, yield and fracture properties of polymeric solids and reinforced polymer composites. Emphasis will be placed on forming techniques for end item fabrication including: extrusion, injection molding, reaction injection molding, thermoforming, and blow molding. The design of end items will be considered in relation to: materials selection, manufacturing engineering, properties, and applications. Prerequisite: MTGN311 or equivalent. 3 hours lecture; 3 semester hours.

MLGN565. MECHANICAL PROPERTIES OF CERAMICS AND COMPOSITES. 3.0 Semester Hrs.
(II) Mechanical properties of ceramics and ceramic-based composites; brittle fracture of solids; toughening mechanisms in composites; fatigue, high temperature mechanical behavior, including fracture, creep deformation. Prerequisites: MTGN445 or MLGN505. 3 hours lecture; 3 semester hours. (Fall of even years only.).

MLGN569. FUEL CELL SCIENCE AND TECHNOLOGY. 3.0 Semester Hrs.
Equivalent with MTGN569,
(II) Investigate fundamentals of fuel-call operation and electrochemistry from a chemical thermodynamics and materials science perspective. Review types of fuel cells, fuel-processing requirements and approaches, and fuel-cell system integration. Examine current topics in fuel-cell science and technology. Fabricate and test operational fuel cells in the Colorado Fuel Cell Center. Prerequisites: EGGN371 or ChEN357 or MTGN351 Thermodynamics I, MATH225 Differential Equations. 3 credit hours.

MLGN570. BIOCOMPATIBILITY OF MATERIALS. 3.0 Semester Hrs.
(II) Introduction to the diversity of biomaterials and applications through examination of the physiologic environment in conjunction with compositional and structural requirements of tissues and organs. Appropriate domains and applications of metals, ceramics and polymers, including implants, sensors, drug delivery, laboratory automation, and tissue engineering are presented. Prerequisites: ESGN 301 or equivalent. 3 hours lecture; 3 semester hours.

MLGN572. BIOMATERIALS. 3.0 Semester Hrs.
Equivalent with MTGN572,
A broad overview on materials science and engineering principles for biomedical applications with three main topics: 1) The fundamental properties of biomaterials; 2) The fundamental concepts in biology; 3) The interactions between biological systems with exogenous materials. Examples including surface energy and surface modification; protein adsorption; cell adhesion, spreading and migration; biomaterials implantation and acute inflammation; blood-materials interactions and thrombosis; biofilm and biomaterials-related pathological reactions. Basic principles of bio-mimetic materials synthesis and assembly will also be introduced. 3 hours lecture; 3 semester hours.

MLGN583. PRINCIPLES AND APPLICATIONS OF SURFACE ANALYSIS TECHNIQUES. 3.0 Semester Hrs.
(II) Instrumental techniques for the characterization of surfaces of solid materials. Applications of such techniques to polymers, corrosion, metallurgy, adhesion science, micro-electronics. Methods of analysis discussed: X-ray photoelectron spectroscopy (XPS), auger electron spectroscopy (AES), ion scattering spectroscopy (ISS), secondary ion mass spectroscopy (SIMS), Rutherford backscattering (RBS), scanning and transmission electron microscopy (SEM, TEM), energy and wavelength dispersive X-ray analysis; principles of these methods, quantification, instrumentation, sample preparation. Prerequisite: B.S. in metallurgy, chemistry, chemical engineering, physics. 3 hours lecture; 3 semester hours. This course taught in alternate even numbered years.

MLGN591. MATERIALS THERMODYNAMICS. 3.0 Semester Hrs.
A review of the thermodynamic principles of work, energy, entropy, free energy, equilibrium, and phase transformations in single and multi-component systems. Students will apply these principles to a broad range of materials systems of current importance including solid state materials, magnetic and piezoelectric materials, alloys, chemical and electrochemical systems, soft and biological materials and nanomaterials. Prerequisite: 300 level thermodynamics, multivariable calculus and differential equations, introductory college chemistry, and introductory materials science courses or consent of instructor.

MLGN592. ADVANCED MATERIALS KINETICS AND TRANSPORT. 3.0 Semester Hrs.
A broad treatment of homogenous and heterogeneous kinetic transport and reaction processes in the gas, liquid, and solid states, with a specific emphasis on heterogeneous kinetic processes involving gas/solid, liquid/solid, and solid/solid systems. Reaction rate theory, nucleation and growth, and phase transformations will be discussed. A detailed overview of mass, heat, and charge transport in condensed phases is provided including a description of fundamental transport mechanisms, the development of general transport equations, and their application to a number of example systems. Prerequisite: 300 level thermodynamics, multivariable calculus and differential equations, introductory college chemistry, and introductory materials science courses or consent of instructor.

MLGN593. BONDING, STRUCTURE, AND CRYSTALLOGRAPHY. 3.0 Semester Hrs.
This course will be an overview of condensed matter structure from the atomic scale to the mesoscale. Students will gain a perspective on electronic structure as it relates to bonding, long range order as it relates to crystallography and amorphous structures, and extend these ideas to nanostructure and microstructure. Examples relating to each hierarchy of structure will be stressed, especially as they relate to reactivity, mechanical properties, and electronic and optical properties. Prerequisite: 300 level thermodynamics, multivariable calculus and differential equations, introductory college chemistry, and introductory materials science courses or consent of instructor.

MLGN597. CASE STUDY - MATERIALS SCIENCE. 0.5-6 Semester Hr.
Individual research or special problem projects supervised by a faculty member.
Course Learning Outcomes

- Graduates will demonstrate the ability to conduct directed research; the ability to assimilate and assess scholarship; and the ability to apply scholarship in new, creative and productive ways.

MLGN598. SPECIAL TOPICS. 0-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

MLGN598. SPECIAL TOPICS. 0-6 Semester Hr.

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MLGN599. CASE STUDY MATERIALS SCIENCE. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree. Prerequisite: Independent Study form must be completed and submitted to the Registrar.

MLGN599. INDEPENDENT STUDY. 0.5-6 Semester Hr.

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MLGN607. CONDENSED MATTER. 3.0 Semester Hrs.
(I) Principles and applications of the quantum theory of electronic in solids: structure and symmetry, electron states and excitations in metals; transport properties. Prerequisite: PHGN520 and PHGN440/MLGN502. 3 hours lecture; 3 semester hours.

MLGN698. SPECIAL TOPICS. 0-6 Semester Hr.
(I, II, S) Pilot course or special topics course. Topics chosen from special interests of instructor(s) and student(s). Usually the course is offered only once, but no more than twice for the same course content. Prerequisite: none. Variable credit: 0 to 6 credit hours. Repeatable for credit under different titles.

MLGN699. INDEPENDENT STUDY. 0.5-6 Semester Hr.
(I, II, S) Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

MLGN699. INDEPENDENT STUDY. 0.5-6 Semester Hr.

MLGN707. GRADUATE THESIS / DISSERTATION RESEARCH CREDIT. 1-15 Semester Hr.
(I, II, S) Research credit hours required for completion of a Masters-level thesis or Doctoral dissertation. Research must be carried out under the direct supervision of the student's faculty advisor. Variable class and semester hours. Repeatable for credit.

Program Directors

Jeramy Zimmerman, Program Director

Michael McGuirk, Associate Director

Department of Chemical and Biological Engineering

Moises Carreon, Professor

Matthew Crane, Assistant Professor

Diego Gomez Gualdron, Assistant Professor

Andrew Herring, Professor

Carolyn Koh, Professor

Melissa Krebs, Associate Professor

Ramya Kumar, Assistant Professor

Stephanie Kwon, Assistant Professor

David Marr, Professor

Joseph Samaniuk, Assistant Professor

Colin Wolden, Professor

David Wu, Professor

Department of Chemistry

Dylan Domaille, Assistant Professor

Tom Gennett, Professor and Department Head

Dan Knauss, Professor

Annalise Maughan, Assistant Professor

C. Michael McGuirk, Assistant Professor

Svitlana Pylypenko, Associate Professor

Ryan Richards, Professor

Alan Sellinger, Professor

Jenifer Shafer, Associate Professor

Brian Trewyn, Associate Professor

Bettina Voelker, Professor

Kim Williams, Professor

David Wu, Professor

Department of Civil and Environmental Engineering

Timothy Strathmann, Professor

Lori Tunstall, Associate Professor

Department of Geology and Geological Engineering

Alexis Navarre-Sitchler, Associate Professor

Department of Mechanical Engineering

Mohsen Asle Zaeem, Professor

Steven DeCaluwe, Associate Professor

Veronica Eliasson, Associate Professor

Joy Gockel, Associate Professor

Owen Hildreth, Associate Professor

Greg Jackson, Professor

Robert Kee, Professor

Leslie Lamberson, Associate Professor

Neal Sullivan, Associate Professor

Brian Thomas, Professor

Xiaoli Zhang, Professor

Department of Metallurgical and Materials Engineering

Gerald Bourne, Teaching Professor

Geoff Brennecka, Associate Professor

Amy Clarke, Professor

Kester Clarke, Associate Professor

Robert Cryderman, Research Associate Professor

Kip Findley, Associate Professor

Brian Gorman, Associate Professor

Megan Holtz, Assistant Professor

Michael Kaufman, Professor

Jonah Klemm-Toole, Associate Professor

Suveen Mathaudhu, Professor

Vladan Stevanovic, Associate Professor

Department of Physics

Tim Ohno, Associate Professor

Meenakshi Singh, Assistant Professor

Eric Toberer, Associate Professor and Program Director

Jeramy Zimmerman, Associate Professor

Professors Emeriti

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Brajendra Mishra, Department of Metallurgical and Materials Engineering

P. Craig Taylor, Department of Physics

Steven Thompson, Department of Metallurgical and Materials Engineering

Chester J. Van Tyne, Department of Metallurgical and Materials Engineering

J. Douglas Way, Department of Chemical and Biological Engineering

Nuclear Engineering

Program Description

The Nuclear Science and Engineering program at Colorado School of Mines is interdisciplinary in nature and draws contributions from departments across the university. While delivering a traditional Nuclear Engineering course core, the School of Mines program in Nuclear Science and Engineering emphasizes the nuclear fuel life cycle. Faculty bring to the program expertise in all aspects of the nuclear fuel life cycle: fuel exploration and processing, nuclear power systems production, design and operation, fuel recycling, storage and waste remediation, radiation detection, and radiation damage as well as the policy issues surrounding each of these activities. Students in all three Nuclear Engineering degrees are exposed to a broad systems overview of the complete nuclear fuel cycle as well as obtaining detailed expertise in a particular component of the cycle. Related research is conducted through the Nuclear Science and Engineering Center with facilities for making advanced measurements on fuel cycle related radioactive materials up to the heaviest reactor-produced element fermium as well as access to the 1 MW GSTR reactor. Students who graduate from the program are prepared for careers in sustainable nuclear energy production, design and analysis of nuclear systems including nuclear reactors, isotope production and purification, national security, or nuclear research.

Student Learning Outcomes

By the end of their nuclear engineering curriculum, students should be able to:

1. Apply fundamentals of nuclear science and engineering in solving open-ended problems

2. Use modern tools of nuclear measurement, analysis, and modeling

3. Locate and use pertinent information from the nuclear literature

4. Interpret and use experimental data for nuclear systems

5. Effectively communicate in both written and oral formats

6. Make responsible contributions to society

Primary Contact

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Program Requirements

The Nuclear Science and Engineering Program offers programs of study leading to three graduate degrees: